CSE 4615: Wireless Networks - Quiz #1 **12 May 2026**
Duration: 20 Minutes **Full Marks: 15**
Student ID:

Let the last two digits of your student ID be denoted by XY.

1. A wireless transmitter sends a signal with power $(XY + 10)$ mW, while the receiver measures a noise power of (XY) μW. 5

   Calculate the Signal-to-Noise Ratio (SNR) in dB and explain whether the communication quality is expected to be reliable or not for wireless communication.

Solution:

Let the last two digits of the student ID be denoted by XY.
Given:

$$S = (XY + 10)\text{ mW}$$

$$N = (XY)\ \mu W$$

Since,

$$1\ \mu W = 10^{-3}\text{ mW}$$

Therefore,

$$N = XY \times 10^{-3}\text{ mW}$$

Hence,

$$\text{SNR} = \frac{S}{N}$$

$$\text{SNR} = \frac{XY + 10}{XY \times 10^{-3}}$$

$$\text{SNR} = \frac{1000(XY + 10)}{XY}$$

SNR in dB:

$$\text{SNR}_{dB} = 10\log_{10}\left(\frac{1000(XY + 10)}{XY}\right)$$

Example for $XY = 25$:

$$S = 35\text{ mW}$$

$$N = 25\mu W = 0.025\text{ mW}$$

$$\text{SNR} = \frac{35}{0.025} = 1400$$

$$\text{SNR}_{dB} = 10\log_{10}(1400) \approx 31.46\text{ dB}$$

Since the SNR is high, the signal is much stronger than noise. Therefore:

- BER becomes lower,
- communication becomes more reliable,
- higher-order modulation can be used.

Thus, the communication quality is expected to be reliable.

2. A mobile user is communicating with a wireless base station while traveling at a speed of $(XY+30)$ km/h. As the user moves through an urban environment, the receiver experiences frequency variations and rapidly changing channel conditions. 5

   Analyze how Doppler shift affects the wireless communication performance in this scenario. Your discussion should include its impact on synchronization, and overall communication reliability.

Solution:

Doppler shift occurs when there is relative motion between the transmitter and receiver in a wireless communication system.

In this scenario, the mobile user is moving at:

$$(XY + 30)\ \text{km/h}$$

As the speed increases, the received signal frequency changes due to the Doppler effect.

Impact of Doppler Shift:

- **Frequency Shift**

  The receiver observes a change in the carrier frequency because the user is moving relative to the base station.
  - Moving toward the base station increases the received frequency.
  - Moving away from the base station decreases the received frequency.

  The maximum Doppler shift is given by:

  $$f_d = \frac{v \cdot f_c}{c}$$

  where v is the mobile speed, f_c is the carrier frequency, and c is the speed of light.
- **Synchronization Problems**

  Wireless receivers are designed to operate at specific carrier frequencies.

  Doppler shift creates:
  - carrier frequency offset (CFO),
  - timing errors,
  - synchronization mismatch between transmitter and receiver.

  As a result, accurate signal detection becomes difficult.
- **Rapid Channel Variation**

  In high-mobility environments, channel conditions change continuously over time.

  This creates:
  - fast fading,
  - unstable signal strength,
  - fluctuating Signal-to-Noise Ratio (SNR).
- **Increase in BER**

  Due to frequency mismatch and unstable channels:
  - symbols may be incorrectly detected,
  - decoding errors increase,
  - Bit Error Rate (BER) becomes higher.
- **Reduced Communication Reliability**

  High BER causes:
  - packet loss,
  - retransmissions,
  - reduced throughput,
  - communication interruptions.

  Therefore, overall communication reliability decreases.

3. A mobile device communicates with a wireless base station using adaptive modulation. Initially, the received SNR is $(XY + 15)$ dB and the device uses QAM256 transmission. As the user moves away from the base station, the SNR decreases by $(XY/2)$ dB due to attenuation, interference, and multipath fading. 5

   Determine the new SNR value and critically analyze whether the system should continue using QAM256, switch to QAM16, or switch to BPSK. Justify your answer using BER, reliability, transmission rate, and wireless channel characteristics.

Solution:

Initial SNR:

$$SNR_i = (XY + 15)\ \text{dB}$$

SNR decrease:

$$\frac{XY}{2}\text{ dB}$$

Therefore,

$$SNR_{new} = (XY + 15) - \frac{XY}{2}$$

Simplifying:

$$SNR_{new} = \frac{XY}{2} + 15$$

Example for $XY = 20$:

$$SNR_i = 35\text{ dB}$$

$$\text{Decrease} = 10\text{ dB}$$

$$SNR_{new} = 25\text{ dB}$$

Modulation Selection:

- **QAM256**
  - Highest throughput
  - Requires very high SNR
  - Sensitive to noise and fading
- **QAM16**
  - Moderate throughput
  - Better robustness
  - Lower BER than QAM256
- **BPSK**
  - Lowest transmission rate
  - Very robust
  - Lowest BER

Decision:

- If SNR is very high: continue using QAM256.
- If SNR becomes moderate: switch to QAM16.
- If SNR becomes low: switch to BPSK.

For the example:

$$SNR_{new} = 25\text{ dB}$$

Therefore:

$$\boxed{\text{QAM16 is the best choice}}$$

because it balances:

- throughput,
- reliability,
- BER performance.

Additional wireless effects:

- **Attenuation:** signal weakens with distance, reducing SNR.
- **Multipath fading:** reflected signals distort the received signal.
- **Doppler shift:** mobility causes frequency changes and synchronization errors.
- **Hidden terminal effect:** collisions increase retransmissions and BER.

Hence, adaptive modulation is necessary to maintain reliable wireless communication.